

ANALYSIS OF FACTORY-TO-CUSTOMER SHIPPING ROUTE EFFICIENCY FOR CANDY DISTRIBUTOR IN NASSAU

SUMMARY

This project is based on analyzing and improving the efficiency of shipping routes from factory to the customer using Data Analytics and Machine Learning techniques. The system will review shipment data, recognize late deliveries, evaluate vendor performance, and project shipment delays. The project enhances logistics planning, transportation efficiency and supply chain decision making.

1. INTRODUCTION

Good transport and logistics management is important to keep supply chain operations running smoothly. Inefficient shipping routes and delivery delays can lead to higher transportation costs and decreased customer satisfaction.

This project studies the shipment data of Nassau Candy Distributor and uses Machine Learning techniques to identify shipment inefficiencies and predict delivery delays.

2. AIMS

- Review shipment performance
- Detect delayed routes
- efficiency of modes of shipment
- Assess vendor performance

3. Objective

- Analyze performance of shipment
- Locate delayed routes

- Evaluate shipment mode efficiency
- Assess vendor performance
- Predict Shipping Delays with Machine Learning
- Enhance logistics management

4. DATASET DESCRIPTION

The dataset includes shipment and transportation data such as:

- Shipment Identification Number
- Location of the factory
- Shipping Method Seller
- Customer Region
- Length
- Time of Transit
- Status of delivery

The dataset processed is:

- Days of Delay
- Efficiency_Score

5. Method

Step 1:

CSV shipment dataset Data Collection.

Step 2:

Data cleaning, null values and duplicate records removal

Step 3:

Exploratory Data Analysis (EDA) of the shipment and vendor analysis

Step 4 :

Feature Engineering for delay_days and efficiency_score.

Step 5:

Training the Linear Regression machine learning model.

Step 6:

Streamlit for visualization and to create dashboards.

6. VISUALIZATION OF DATA

The following visualizations have been created:

- Chart of Shipment Mode Analysis
- Pie Chart of Delivery Status
- Vendor Performance Graph

These visualizations assist in understanding logistics patterns and shipment effectiveness.”

7. AUTOMATED LEARNING MODEL

Used Linear Regression algorithm to predict shipment delays using:

- Shipping Cost
- Range

The model supports better planning of transportation and helps in detecting possible shipment delays.

8. RESULTS

- identified delayed shipment routes

- Monitored vendor ship performance
- Visualized distribution of delivery status
- Shipment delay predictions were made
- Route efficiency analysis complete

9. PROJECT NOVELTY

Unlike existing shipment tracking systems, this project includes:

- Delay prediction using machine learning
- Intelligent route efficiency analysis
- Track vendor performance
- Insights into logistics optimization

10. FUTURE WORK

- Shipment tracking in real-time
- Route optimization with GPS
- Advanced Machine Learning algorithms• Deploying the dashboard to the Cloud

11. Conclusions

In this project, shipment routes have been successfully analyzed and the delivery delays have been predicted using Data Analytics and Machine learning techniques. The system improves logistics efficiency, provides better decision making and assists in reducing transportation delays for Nassau Candy Distributor.

11. BIBLIOGRAPHY

- Python Docs
- Pandas Docs
- Documentation of Scikit-learn
- Streamlit docs
- Research Papers on Logistics & Supply Chain